

MAHDI SAMÈ

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EDUCATION

University of Tehran

Bachelor of Science in Mechanical Engineering

Tehran, Iran

Sep 2019 - Jul 2024

- GPA: 3.84/4.0 (18.08/20)

RESEARCH INTERESTS

- | | |
|---------------------------|----------------------|
| ✓ Robotics | ✓ Controls |
| ✓ Mechatronics | ✓ Biomedical Devices |
| ✓ Human-Robot Interaction | ✓ Machine Learning |

PUBLICATION

- | | |
|--|----------|
| • SM. Same , EB. Sarmadi, M. Abolghasemi, A. Bahrami, MR. Abolghasemi, AY. Koma
“ <i>Design and Fabrication of a Microinjector to Conduct Columnar Injection of Stem Cells for Treating Parkinson’s Patients</i> ” ICRoM 2024, 12th RSI International Conference | Dec 2024 |
| • A. Naeini, SM. Same , J. Khorramdel, A. Ghanbarzadeh, R. Behbahaninejad, AY. Koma
“ <i>DILP: Diverse Iranian License Plate Detection and Recognition Dataset</i> ” ICRoM 2024, 12th RSI International Conference | Dec 2024 |
| • S. Sehhat, AY. Koma, SM. Same , P. Abdollahnezhad “ <i>Online Stabilization of Ankle Prosthesis on Uneven Surfaces</i> ” ICRoM 2022, 10th RSI International Conference | Nov 2022 |

RESEARCH EXPERIENCE

Convergent Technologies Research Center, Research Assistant

Feb 2023 - Jul 2024

Website: <https://utnbic.ir/en>

- **Automatic Injector Microdrive**, (bachelor’s degree thesis)
 - Designed and fabricated an automated injector microdrive to precisely administer medication into Parkinson’s patients’ brains, implementing controllers to improve treatment efficiency
- **Virtual Reality Device for Rodents**
 - Designed and built an advanced virtual reality device with improved comfort by minimizing friction
 - Developed an automated reward system to streamline rat training processes in behavioral research

Center of Advanced Systems and Technologies (CAST), Research Assistant

Dec 2021 - Present

Website: <https://cast-tech.ir>

- **Surena Humanoid Robot** (Choreonoid and ROS), Ongoing
 - Contributing to the Dynamics and Control team, gaining expertise in robot dynamics and kinematics
 - Developing controllers and code files and initiating robot simulations in Choreonoid to support ongoing trajectory planning projects
- **Control Position of Ankle Prosthesis on Flat Surfaces** (Python and Arduino)
 - Designed a data acquisition system for ankle prostheses, collected diverse obstacle negotiation data, and developed a stabilizing controller for ascending and descending obstacles

TEACHING EXPERIENCE

- | | | |
|---|--------------------|-----------|
| • Measurement Systems and Laboratory | Teaching Assistant | Fall 2023 |
| • Optimization of Mechanical Systems | Teaching Assistant | Fall 2023 |

• Design of Machine Elements II	Teaching Assistant	Spring 2023
• Mechanical Vibrations	Teaching Assistant	Fall 2022
• Engineering Mathematics	Teaching Assistant	Spring 2022

SELECTED COURSES

- | | |
|---------------------------------------|--|
| ✓ Mechatronics (4/4) | ✓ Machine Learning (4/4) |
| ✓ Automatic Control (4/4) | ✓ Optimization of Mechanical Systems (4/4) |
| ✓ Circuit and Electric Machines (4/4) | ✓ Characterization in Nano Engineering |

SKILLS

- | | | |
|-------------------------|------------------------------|--------------------------|
| • Programming: | • Simulation and CAD: | • Others: |
| ✓ Python (Professional) | ✓ Choreonoid (Intermediate) | ✓ ROS (Intermediate) |
| ✓ CPP (Professional) | ✓ SolidWorks (Professional) | ✓ Arduino (Professional) |
| ✓ MATLAB (Professional) | ✓ ADAMS (Professional) | ✓ Git (Intermediate) |
| ✓ C# (Intermediate) | ✓ ANSYS (Intermediate) | ✓ LaTeX (Intermediate) |

SELECTED PROJECTS

- **Self-balancing Robot** (MATLAB, Arduino & SolidWorks) Spring 2023
 - Designed a two-wheeled robot using SolidWorks and developed motion stabilization controllers using Simscape and MATLAB Control System Toolbox
 - Constructed the robot and successfully implemented the designed controllers for real-world applications
- **Inline Load Cell** (ANSYS and Arduino) Spring 2023
 - Designed an Inline load cell in SolidWorks and conducted structural analysis in ANSYS
 - Fabricated the load cell and used Arduino for precise small weight measurements
- **Speech Emotion Recognition** (Python) Spring 2022
 - Compiled a dataset featuring diverse voice recordings, including both genders and various emotions
 - Applied clustering and classification algorithms to assess and compare their performance in emotion recognition and voice analysis

AWARDS & HONORS

- Ranked among the top 15% of the class of 2023 in the School of Mechanical Engineering
- Ranked among the top 0.3% in the Iranian university entrance exam

LANGUAGES

- ✓ English (Professional working proficiency)
 - **IELTS** Band Score: 8.0 (Reading: 9.0 , Listening: 8.5 , Writing: 6.5 , Speaking: 7.0)
- ✓ Spanish (Intermediate proficiency)
- ✓ Persian (Native proficiency)

REFERENCES

- | | | |
|---------------------------------|---|------------------------|
| • Dr. Farshad Kowsary | Professor, University of Tehran | fkowsari@ut.ac.ir |
| • Dr. Aghil Yousefi-Koma | Professor, University of Tehran | aykoma@ut.ac.ir |
| • Dr. Arash Bahrami | Assistant Professor, University of Tehran | arash.bahrami@ut.ac.ir |